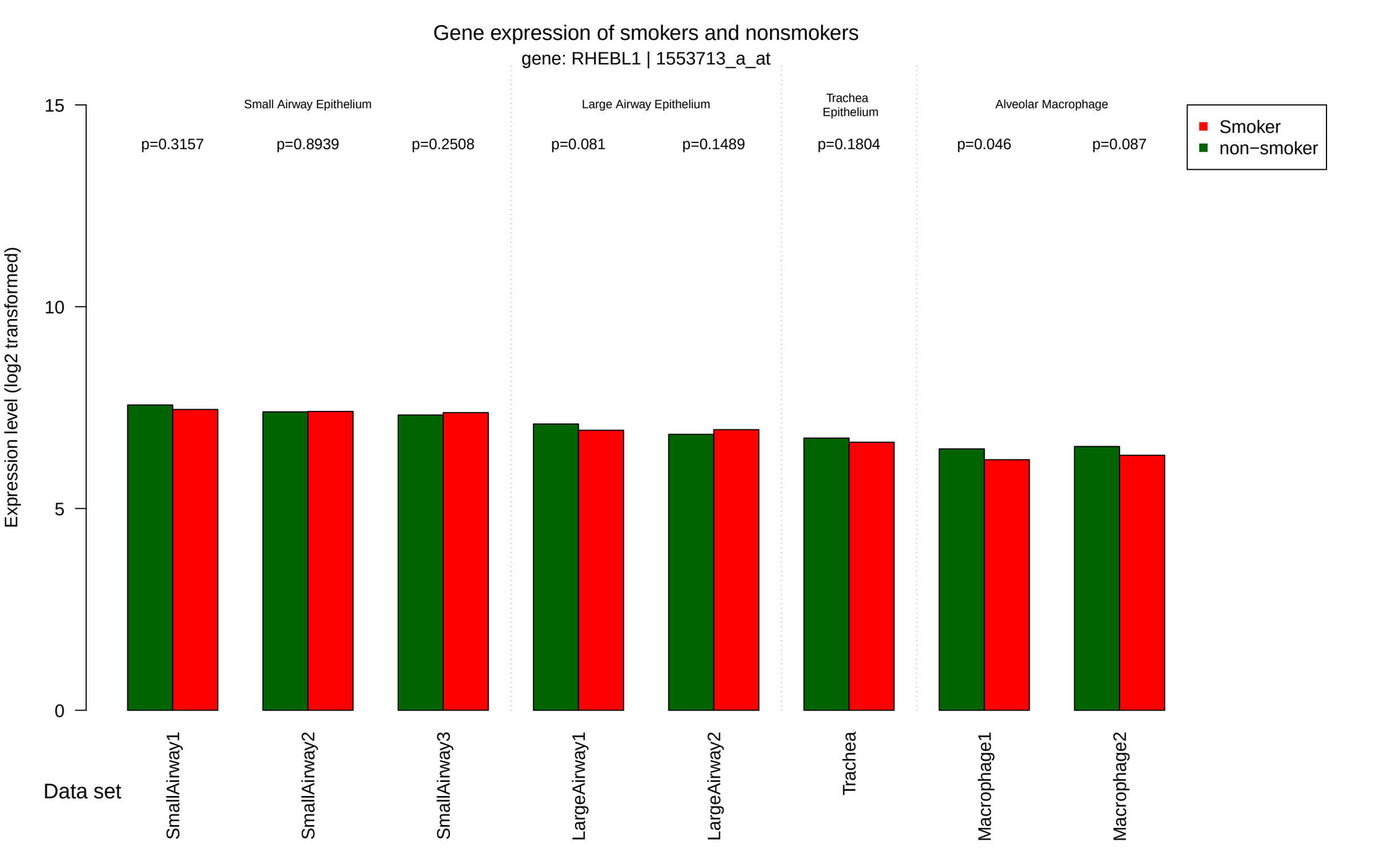
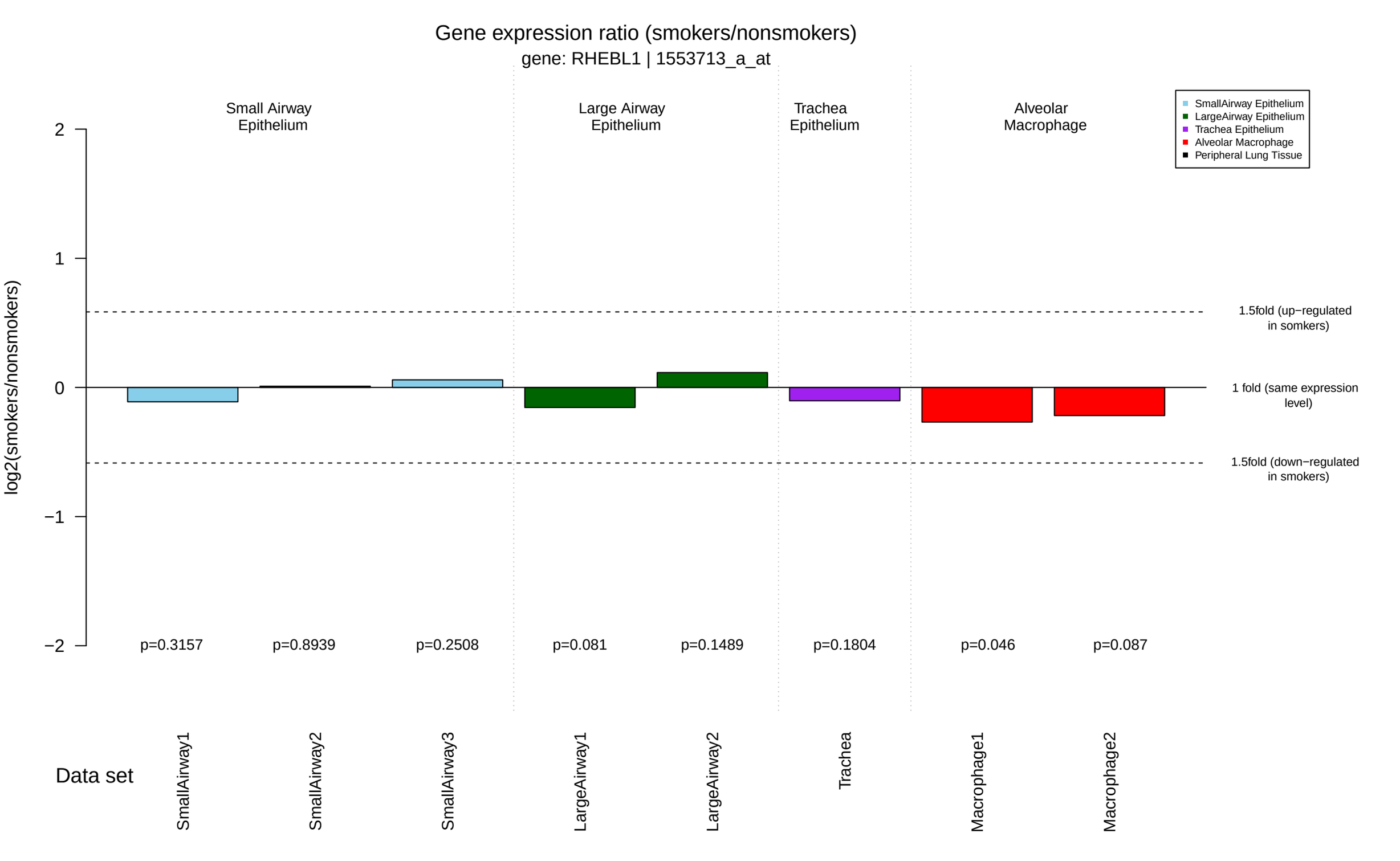
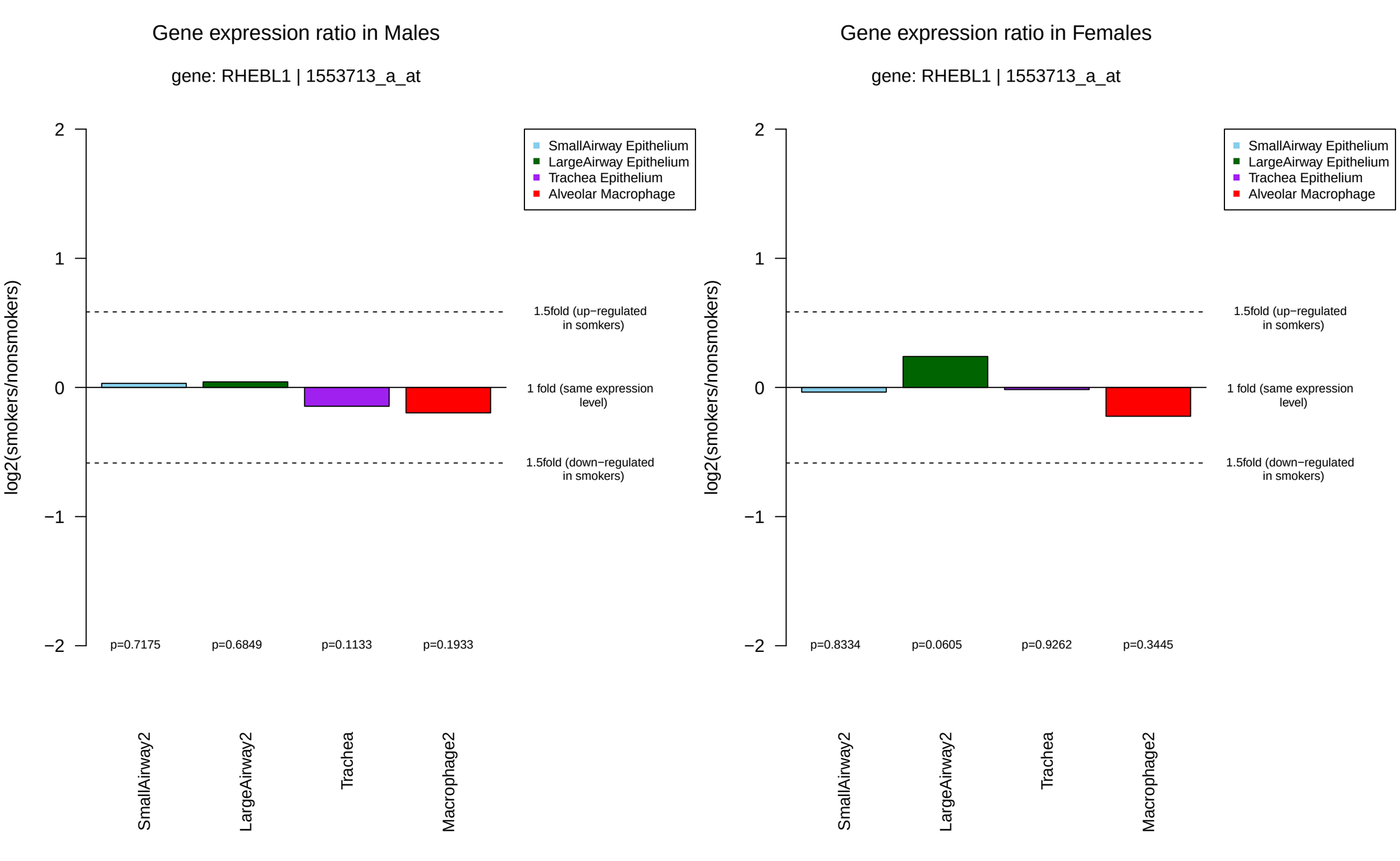
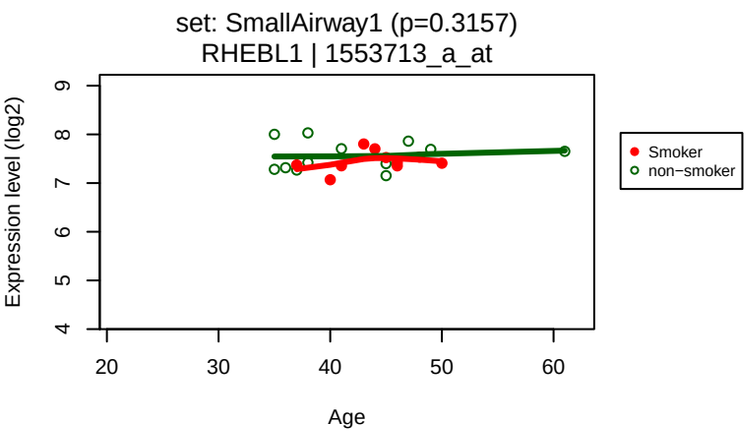
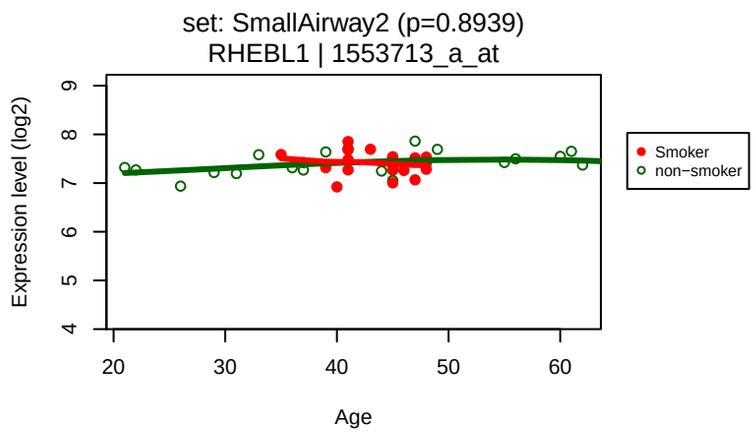
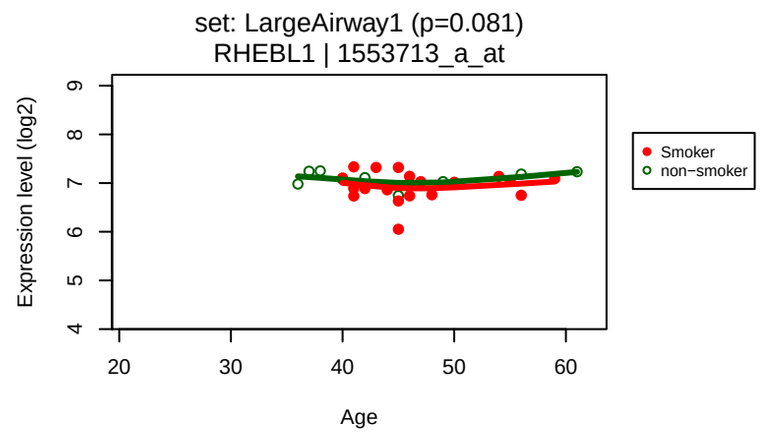

Gene expression of smokers and nonsmokers
gene: RHEBL1 | 1553713_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3157
p=0.8939
p=0.2508
p=0.081
p=0.1489
p=0.1804
p=0.046
p=0.087
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

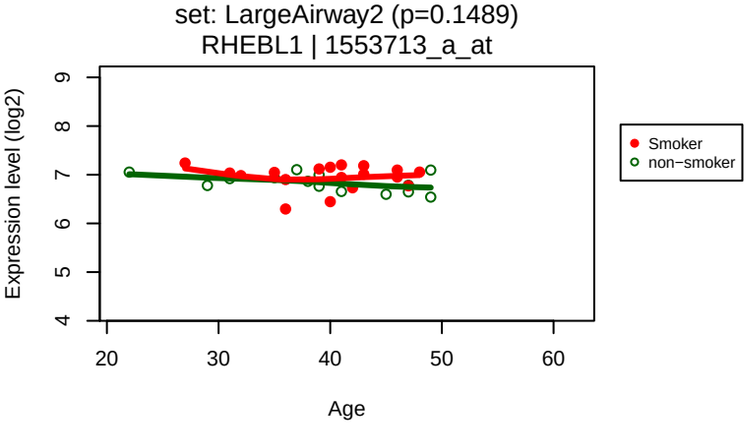

Gene expression ratio (smokers/nonsmokers)
gene: RHEBL1 | 1553713_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3157
p=0.8939
p=0.2508
p=0.081
p=0.1489
p=0.1804
p=0.046
p=0.087
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

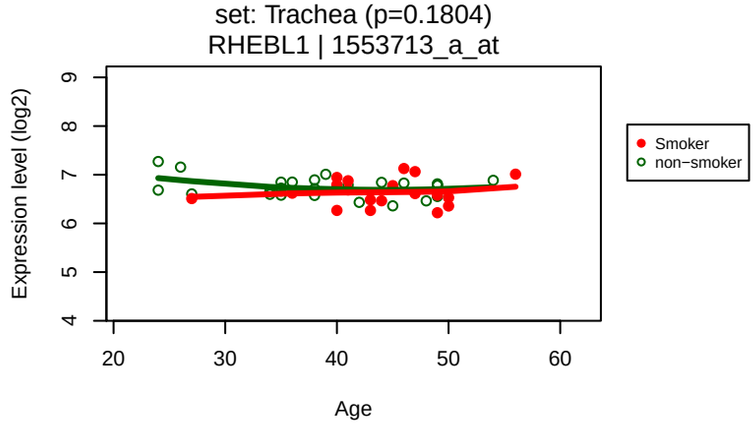

Gene expression ratio in Males
gene: RHEBL1 | 1553713_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7175
p=0.6849
p=0.1133
p=0.1933
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHEBL1 | 1553713_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8334
p=0.0605
p=0.9262
p=0.3445
SmallAirway2
LargeAirway2
Trachea
Macrophage2

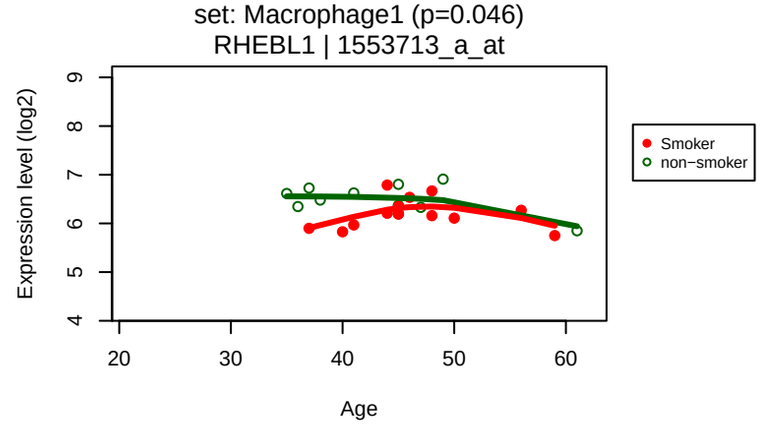

set: SmallAirway1 (p=0.3157)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker

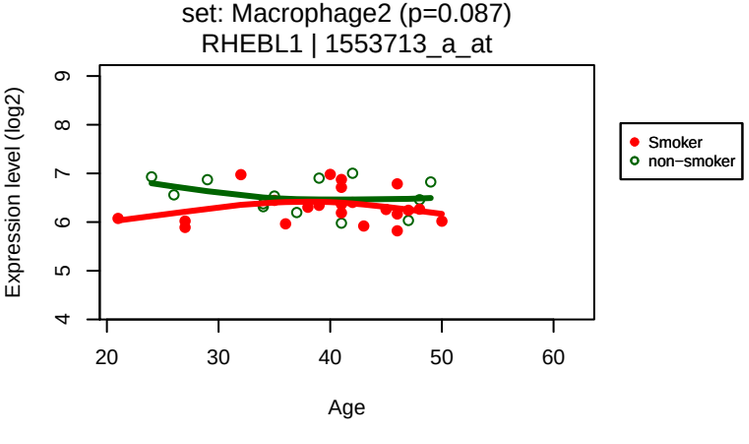

set: SmallAirway2 (p=0.8939)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker

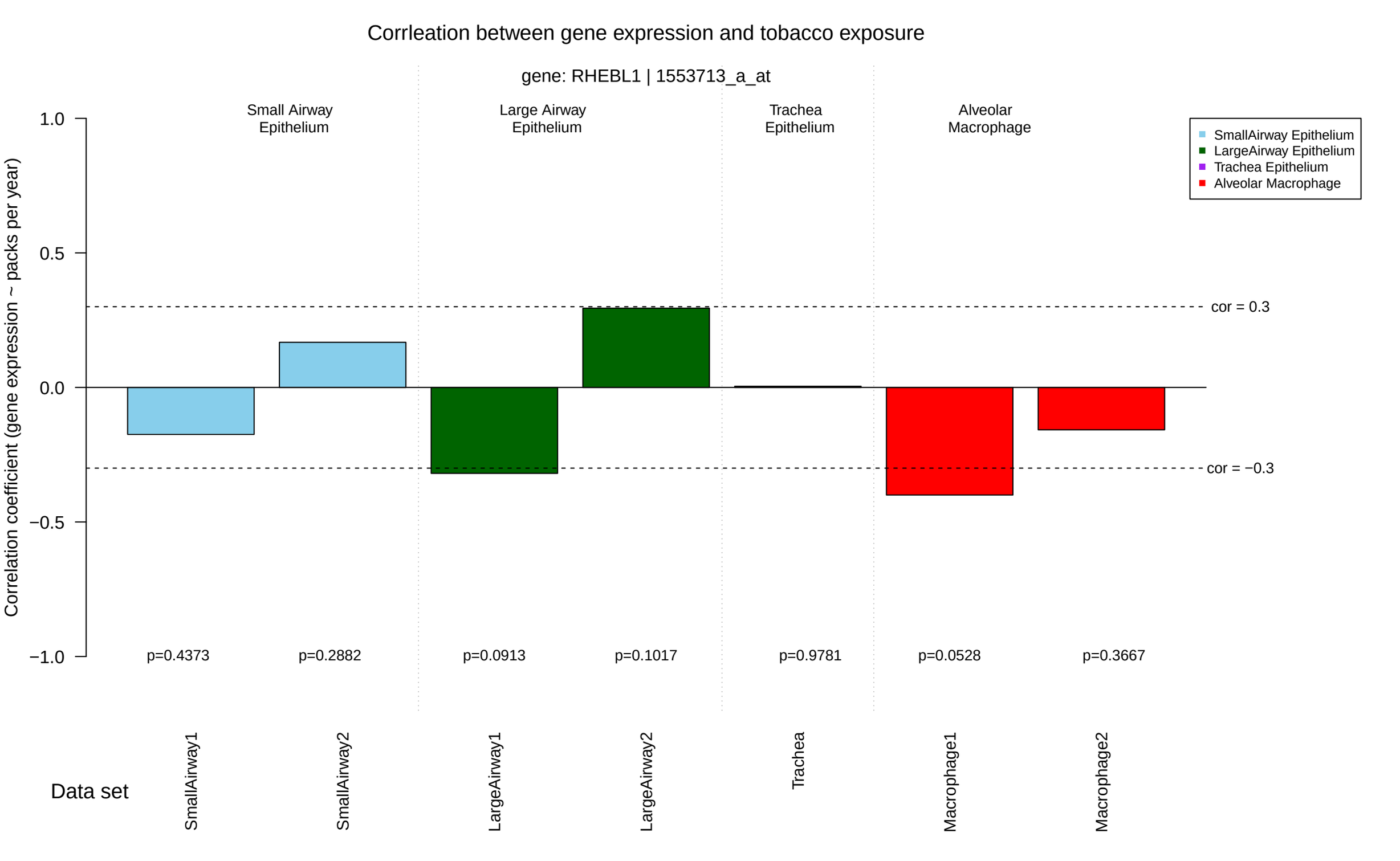

set: LargeAirway1 (p=0.081)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1489)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1804)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.046)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.087)
RHEBL1 | 1553713_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHEBL1 | 1553713_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4373
p=0.2882
p=0.0913
p=0.1017
p=0.9781
p=0.0528
p=0.3667
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2